



China's rocket program

China is expanding its rocket launch capabilities with a new series of rockets including super-heavy lift and reusable rockets. <https://digit.in/Aug18-20-1>



AI defeats humans

DeepMind AI playing a stripped down version of "Capture the flag" in Quake III exceeded "human-level" gameplay. <https://digit.in/Aug18-20-2>

EVERYTHING WE KNOW ABOUT THE NVIDIA TURING GTX 1180 / 1170 / 1160

Little is known about NVIDIA's next gen GPU so we decided to get our hands dirty and find out everything we could about the new GPUs.

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here has been plenty of speculation about what the next generation of graphics cards from NVIDIA are

going to bring to the table. We've heard of a few rumours and there certainly have been a couple of leaks of late that have let us postulate the specifications and prowess that these new cards will sport. Do note that these speculations that we're going to dive into are based off unreleased material, leaked benchmarks and NVIDIA's technology history. So we're going to be off the mark but not by a large margin.

WHAT WILL NVIDIA'S NEXT GPU ARCHITECTURE BE CALLED?

NVIDIA unveiled Volta on December 7, 2017 in the form of the NVIDIA TITAN V at the NIPS technology conference. Which was based on the more efficient 12nm fabrication process with 12 GB of HBM2 memory. With the last two generations, i.e. Maxwell and Pascal, NVIDIA kept the architecture name the same as when they first introduced them. However, rumours started that the upcoming cards might be called Turing or Ampere. All of that was set aside when

a Forbes story about NVIDIA's financial results mentioned that the upcoming gaming graphics cards from NVIDIA will be codenamed Turing and not Volta.

It is extremely unlikely that Turing is an entirely different architecture from ground up. Turing might be what NVIDIA is calling the gaming variants of Volta.

NVIDIA GTX 1180 / 1170 / 1160 OR GTX 2080 / 2070 / 2060 ?

NVIDIA Turing cards are either going to be GTX 2000 series or GTX 1100 series. Why this confusion? Because this is the first time a 4-digit numbered graphics card is getting a successor. Till now, we had the 3-digit nomenclature which had the newer generations increasing the first digit i.e. GTX 400 made way to the GTX 500 which then led to GTX 600, with the missing GTX 800 series being the only exception, since NVIDIA already had GTX 800M mobile chips.

Pascal returned to the typical naming scheme by changing the 9 to a 10. So came the GTX 10xx (1080/1070/1060/...) series that we now see everywhere. NVIDIA Turing based graphics cards are expected to follow in a similar fashion and launch as GTX 1100 series i.e. GTX 1180 / 1170

/ 1160. We've heard the same from two industry sources. There was also a recent E3 2018 video interview with YouTuber Brainbean where a Lenovo representative let it slip that the GTX 11-series cards were slated to come later this (2018) fall.

NVIDIA TURING GTX 1180 / 1170 / 1160 RELEASE DATE

We now know of a release window in the Aug-Sep period. Secondly, Tom's Hardware Germany heard from sources that NVIDIA has begun briefing board partners back in early June 2018. They even managed to obtain a Bill of Materials (BoM) from NVIDIA to their partners which details all the stages of a card's development cycle. The time period as per this BoM adds up to roughly ten weeks. With this leak from mid-June, that works out to an Aug-Sep release.

We've also heard from an industry source that the GTX 1180 will be unveiled at Gamescom. Additionally, the schedule for the Hot Chips symposium initially had (since corrected) a session titled "NVIDIA's Next Generation Mainstream GPU" to be presented by Stuart Oberman, Principal Engineer, NVIDIA. If NVIDIA was going to detail its next-gen